

NEHA SHIRODKAR

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SUMMARY

Software Engineer (GenAI & Backend) with 5+ years delivering production systems across fintech, data platforms, and high-volume SaaS. At Intuit, architected and scaled multi-agent AI services on GenOS handling **200K+** daily requests using **Python**, **FastAPI**, **LangGraph**, **OpenAI APIs**, and **AWS Bedrock**, improving RAG answer accuracy by **31%** and saving analysts **4,000+** hours of manual review per year. Previously built large-scale customer and commerce platforms on **Java**, **Spring Boot**, and **Django**, handling tens of millions of records across batch and streaming pipelines. Strong end-to-end owner of backend and agentic AI systems from **LLMOps** and multi-agent orchestration to distributed microservices delivering outcomes like **sub-10-minute** operational turnaround, **34%** reduction in analyst effort, and **99.9%** production availability.

EXPERIENCE

INTUIT (QUICKBOOKS)

Jan 2025 – Present

Software Engineer - GenAI & Backend

Remote, US

- Scaled **agentic AI** backend services in **Python** and **FastAPI** across distributed microservices to handle **200K+** daily requests, automating case analysis and routing that saved analysts **4,000+ hours** of manual review per year.
- Built an LLM case summarization pipeline on Intuit's **GenOS** platform using **OpenAI** and **Claude** via **AWS Bedrock**, cutting investigation time from **3–4 hours** per case to under **10 minutes** across recurring operational queues.
- Architected multi-agent workflows using **LangGraph** with parallel planner, retrieval and tool-calling nodes; implemented **hybrid retrieval (BM25 + RRF reranking)** to improve RAG answer accuracy by **31%** and reduce analyst escalations by **800+** per month.
- Built a **RAG pipeline** using **LangChain** and **FAISS** grounded in **500+** QuickBooks policy documents, achieving **91%** factual accuracy on edge-case prompts; exposed **FastAPI** endpoints as callable agent tools in GenOS (GenRuntime) via function-calling patterns.
- Delivered end-to-end backend reliability for **5** production FastAPI services, leading incident response and **PostgreSQL** schema migrations across **40M+** records with zero data loss; enforced API versioning across **6+** internal teams via **PyTest** contract suites.
- Instrumented inference endpoints with **LangSmith** tracing and **RAGAS** for live A/B testing and golden-set benchmarking; deployed **GenAI services** on **AWS (ECS, Lambda, S3, CloudWatch)** maintaining **99.9%** availability through zero-downtime releases via **GitHub Actions** and **Docker**.
- Authored prompt engineering templates through **spaCy**-based context structuring to pre-process financial attributes before LLM calls, cutting vague or incomplete model responses by **22%** across test query sets.

INFINITE INFOLAB

Aug 2021 – Jul 2023

Software Engineer

Hybrid, India

- Engineered a multi-tenant Customer Data Platform on **Java** and **Spring Boot**, consolidating **40M+** records from billing, CRM, and transaction systems into a unified queryable layer for enterprise lending and payments clients.
- Designed real-time data sync pipelines using **Apache NiFi** and **Kafka**, cutting downstream report delays from same day to near real-time and saving teams **6+** hours of lag per reporting cycle.
- Optimized a **MySQL** customer schema with partition-aware indexing, dropping query latency from **900ms to under 200ms** and enabling sub-second lookups for credit segmentation across **40M+** records.
- Secured multi-tenant REST APIs with **OAuth 2.0**, **JWT**, and **RBAC**; enforced quality through **JUnit5** and **Testcontainers**, raising test coverage to **82%** and cutting defect rate by **29%**; directed CI pipelines through **Jenkins** with **SonarQube** analysis, reducing production defects by **28%**.
- Exposed a normalized customer data layer through versioned **Spring Boot REST APIs** consumed by internal marketing and analytics dashboards, improving campaign targeting turnaround by **29%** and eliminating ad-hoc data pull requests from business teams.

VIVMA SOFTWARE INC.

Aug 2019 – Jul 2021

Software Development Engineer

Remote, India

- Developed **Python** and **Django** backend services for a multi-client payments platform, automating order routing logic that expanded peak transaction processing capacity by **34%** without adding infrastructure.
- Implemented async payment processors with **Celery** and **RabbitMQ** with retry policies and timeout handling for fault-tolerant order processing, eliminating manual status tracking and cutting end-to-end order cycle time from **4 hours to under 2.5 hours** across concurrent client workloads.
- Architected **MongoDB** document storage for high-write payment workloads, improving transaction durability by **42%** during seasonal demand spikes with no increase in p99 latency.
- Delivered **GraphQL** APIs for **8+** partner integrations, reducing payload size by **36%**; containerized services with **Podman** and built **Bitbucket Pipelines** with automated **PyTest** suites, reducing regression defects by **31%**.

PROJECTS

MCP-Powered AML/KYC Compliance Agent

May – Present

Python, LangGraph, MCP (Model Context Protocol), LangSmith, ChromaDB, FastAPI

[GitHub](#)

- Built an agentic compliance assistant using **MCP** and **LangGraph** exposing 3 financial compliance tools (entity resolution, AML risk scoring, BSA clause lookup) to a **Claude Sonnet** agent via **Anthropic API**, automating regulatory lookups to under seconds per query.
- Improved citation accuracy from **42% to 96%** through iterative **LangSmith-driven prompt and retrieval tuning**, cutting per-query cost by **39%**.
- Implemented **human-in-the-loop** escalation with confidence-thresholded routing, an audit trail, and **48 PyTest cases** for production-grade reliability.

Mental Health Discourse Analyzer

Jan 2026 – Feb 2026

Python, RoBERTa (fine-tuning), BERTopic, FastAPI, MLflow, Hugging Face Spaces

[GitHub](#)

- **Fine-tuned RoBERTa** on 7.8K Reddit posts to classify mental health distress severity, achieving **0.91** accuracy and **0.90** macro F1 despite a 3:1 class imbalance.
- Applied **BERTopic** for unsupervised theme extraction across mental health communities; tracked all experiments via **MLflow** and deployed a live **Streamlit** dashboard on **Hugging Face Spaces**.
- Designed the model to generalize from post content alone, never seeing subreddit names to avoid memorizing weak-label source cues.

Clinical Re-Admission Risk Predictor

Oct 2025 – Dec 2025

Python, XGBoost, SHAP, scikit-learn, FastAPI, MLflow, Hugging Face Spaces

[GitHub](#)

- Built a 30-day hospital readmission predictor on **100K+** patient encounters using **XGBoost** and **SMOTE**, achieving **AUC 0.64** consistent with the published ceiling for this dataset.
- Surfaced per-patient top 5 risk drivers using **SHAP explainability** and tuned the decision threshold to a clinical rule thereby maximizing precision with **recall $\geq$ 0.50**.
- Deployed a live **FastAPI + Streamlit** dashboard on **Hugging Face Spaces** with **MLflow** experiment tracking and honest evaluation methodology - no data leakage, no inflated metrics.

TECHNICAL SKILLS

Languages & Frameworks: Python (FastAPI, Django, Flask), Java (Spring Boot, JUnit), JavaScript (Node.js), TypeScript, SQL
GenAI & Agentic AI:

- **Orchestration & Agents:** LangGraph, LangChain, LlamaIndex, CrewAI, MCP (Model Context Protocol), GenOS (GenRuntime, GenOrchestrator)
- **RAG & Retrieval:** RAG pipelines, hybrid retrieval (BM25, RRF reranking), semantic chunking, reranking
- **Vector Databases:** Pinecone, ChromaDB, FAISS
- **LLM & Tooling:** OpenAI APIs, AWS Bedrock (Claude), LangSmith (observability & tracing), RAGAS (LLM evaluation), Guardrails AI, spaCy
- **Techniques:** Prompt engineering, context engineering, tool calling / function calling, HITL escalation, structured outputs (Pydantic)

ML & Data Science: PyTorch, XGBoost, scikit-learn, Hugging Face, BERT / RoBERTa / LLaMA (fine-tuning), MLflow, Weights & Biases, Pandas, NumPy

Cloud & DevOps: AWS (ECS, Lambda, S3, RDS, CloudWatch, API Gateway), Docker, Kubernetes, GitHub Actions, Jenkins, SonarQube

Data & Backend: PostgreSQL, MySQL, MongoDB, Redis, Kafka, RabbitMQ, Celery, Apache NiFi, GraphQL, gRPC, RESTful APIs, OAuth 2.0, JWT, RBAC

Practices: PyTest, JUnit5, Testcontainers, CI/CD, A/B evaluation, zero-downtime deployments, Agile/Scrum, HIPAA/FHIR/DICOM, PCI-DSS

EDUCATION

Master of Science | Information Systems | Syracuse University, New York

May 2025

Bachelor of Engineering | Computer Science | University of Mumbai, Mumbai

June 2021